$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \qquad \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \qquad \begin{pmatrix} a_{11} & \cdots & a_{1k} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2k} & \cdots & a_{2n} \\ \vdots & \ddots & a_{3k} & \cdots & \vdots \\ a_{n1} & \cdots & a_{nk} & \cdots & a_{nn} \end{pmatrix}$$

$$\underset{(q^n - 1)\ \textbf{ways}}{\mathbf{v}_1 \neq \mathbf{0}} \times \underset{(q^n - q)\ \textbf{ways}}{\mathbf{v}_2 \notin \textbf{span}(\mathbf{v}_1)} \times \cdots \times \underset{(q^n - q^{k-1})\ \textbf{ways}}{\mathbf{v}_k \notin \textbf{span}(\mathbf{v}_i)_{i=1}^{k-1}} = \prod_{i=0}^{k-1}(q^n - q^i)$$

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix} \qquad \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix} \qquad \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix}$$

$$\underset{(q^k - 1)\ \textbf{ways}}{\mathbf{w}_1 \in W \setminus \{0\}} \qquad \underset{(q^k - q)\ \textbf{ways}}{\mathbf{w}_2 \notin \textbf{span}(\mathbf{v}_1)} \times \cdots \times \underset{(q^k - q^{k-1})\ \textbf{ways}}{\mathbf{w}_k \notin \textbf{span}(\mathbf{w}_i)_{i=1}^{k-1}} = \prod_{i=0}^{k-1}(q^k - q^i)$$